Continuous Functions

Definition

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if the preimage of every open set in Y is open in X.

f is continuous at $x \in X$ if the preimage of every open set containing $f(x)$ contains an open set containing x.

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if f is continuous at every point $x \in X$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $f^{-1}(F)$ is closed in X for every closed set F in Y.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for every $A \subseteq X$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$ for every $B \subseteq Y$.

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Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $f^{-1}(\mathring{B}) \subseteq (f^{-1}(B))^{\circ}$ for every $B \subseteq Y$.

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f is continuous if and only if $f(\mathring{A}) \subseteq (f(A))^{\circ}$ for every $A \subseteq X$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $f^{-1}(\partial B) \subseteq \partial(f^{-1}(B))$ for every $B \subseteq Y$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $\partial(f(A)) \subseteq f(\partial A)$ for every $A \subseteq X$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $f(\mathrm{Int}(A)) \subseteq \mathrm{Int}(f(A))$ for every $A \subseteq X$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $\mathrm{Int}(f^{-1}(B)) \subseteq f^{-1}(\mathrm{Int}(B))$ for every $B \subseteq Y$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $f^{-1}(\mathrm{Ext}(B)) \subseteq \mathrm{Ext}(f^{-1}(B))$ for every $B \subseteq Y$.

Theorem

Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $\mathrm{Ext}(f(A)) \subseteq f(\mathrm{Ext}(A))$ for every $A \subseteq X$.

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Let $f: X \to Y$ be a function between topological spaces X and Y.

f is continuous if and only if $f^{-1}(\mathrm{Ext}(B)) = \mathrm{Ext}(f^{-1}(B))$ for every $B \subseteq Y$.

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